Timothy J. Aveni

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Education

Aug. 2022-present **University of California, Berkeley** *Berkeley, CA*
Ph.D. student, Computer Science GPA: 4.0
Human-Computer Interaction

Aug. 2016-May 2019 **Georgia Institute of Technology** *Atlanta, GA*
Bachelor of Science in Computer Science GPA: 4.0
Specializations: Artificial intelligence, Information and networking

Research experience

Aug. 2022-present **Berkeley Institute of Design**, University of California, Berkeley *Berkeley, CA*
Ph.D. student *Advised by Björn Hartmann, Armando Fox*

- Studying human-computer interaction, especially as it relates to how people instruct their computers and how we can teach people to better construct digital workflows
- Building tools to enable faster prototyping of AI-backed tools, both for skilled and for novice programmers

Nov. 2016-May 2019 **Contextual Computing Group**, Georgia Institute of Technology *Atlanta, GA*
Undergraduate Researcher *Advised by Caitlyn Seim under Thad Starner*

- Explored the acquisition of computer stenography skills through passive haptic stimuli
- Built hardware, firmware, and software, start to finish, to power and analyze ~50 user trials
- Built Android app to train motor skills, used in "Towards Haptic Learning on a Smartwatch"

Publications

Peer-reviewed

2026 J. Smith, S. G. Almeda, T. J. Aveni, A. Agarwal, and B. Hartmann. 2026. "Noise Pilot: Enabling Artistic Workflow Composition with Diffusion-Based Image Generation". In proceedings of *CHI '26*. *In press.*

2025 T. J. Aveni, H. Mor, A. Fox, and B. Hartmann. 2025. "Generative Trigger-Action Programming with Ply". In proceedings of *UIST '25*.

2025 T. J. Aveni, J. Smith, A. Fox, and B. Hartmann. 2025. "Supporting Students in Prototyping AI-backed Software with Hosted Prompt Template APIs". In proceedings of *ITiCSE '25*.

2023 T. J. Aveni, A. Fox, and B. Hartmann. 2023. "Bringing Context-Aware Completion Suggestions to Arbitrary Text Entry Interfaces". Demo, in adjunct proceedings of *UIST '23*.

2019 T. J. Aveni, C. Seim and T. Starner, "A preliminary apparatus and teaching structure for passive tactile training of stenography," In proceedings of *2019 IEEE World Haptics Conference (WHC)*.

2018 C. Seim, R. Pontes, S. Kadiveti, Z. Adamjee, A. Cochran, T. Aveni, P. Presti, T. Starner. "Towards Haptic Learning on a Smartwatch," Poster, in proceedings of *ISWC '18*.

Additional scholarship

2023 T. J. Aveni, A. Fox, and B. Hartmann. 2023. "OmniFill: Domain-Agnostic Form Filling Suggestions Using Multi-Faceted Context". *arXiv:2310.17826 [cs.HC]*.

2019 T. J. Aveni, "Passive Haptic Learning for Computer Stenography," 2019. Undergraduate Thesis.

Keywords human-computer interaction, human-centered AI, LLMs, end-user programming, AI agents

Teaching experience

Spring 2026 **Graduate Student Instructor (TA)** and **Software Lead**, Foundations of Data Science *Berkeley, CA*

- Help to develop, deploy, and integrate Jupytutor, an AI support tutor that lives in Jupyter notebooks for data science courses
- Hold weekly discussion and lab section

Fall 2025, Spring 2025, Spring 2024 **Graduate Student Instructor (TA)**, User Interface Design and Development *Berkeley, CA*

- Held weekly discussion sections
- Developed reagent, a platform to facilitate rapid prototyping of AI-backed software
- Designed new homework to introduce students to developing intelligent user interfaces

Summer 2025, Summer 2024 **Instructor**, User Interface Design and Development *Berkeley, CA*

- Developed and delivered course lectures for two summer classes
- Developed and deployed a platform for interactive presentations of course material
- Reworked homeworks to enable targeted automatic feedback

Fall 2023 **Graduate Student Instructor (TA)**, Introduction to Software Engineering *Berkeley, CA*

- Led a team of 7 teaching assistants
- Held two weekly discussion sections

Aug. 2020-May 2021 **Volunteer teacher**, Introduction to Computer Science *Ajo, AZ (remote)*

TEALS Program (Technology Education and Literacy in Schools), Microsoft Philanthropies

- Taught an introductory high school computer science course in a team with one other volunteer teacher, two volunteer TAs, and a classroom teacher
- Led two classes per week remotely

Aug. 2018-May 2019 **Head Teaching Assistant**, Data Structures and Algorithms (CS 1332), Georgia Institute of Technology

- Managed ~30 teaching assistants in a class with ~600 students per semester
- Developed and taught various lectures, designed exam material, oversaw exam grading
- Organized hiring and interviewing for new TAs

Jan. 2017-May 2018 **Teaching Assistant**, Data Structures and Algorithms (CS 1332), Georgia Institute of Technology

- Held weekly recitation and office hours
- Graded and provided feedback on Java homework assignments and exams
- Developed homework grading assistant software used by all CS 1332 TAs

Industry experience

Aug. 2020-June 2022 **Senior Software Engineer**, Gradescope *Oakland, CA*

Turnitin

- Wrote full-stack Rails/React code for products that help instructors grade assessments
- Remediated accessibility issues to meet WCAG 2.1 recommendations
- Introduced static code analysis and automation to improve i18n workflows
- Participated in the user-facing engineering email support rotation

June 2019-June 2020 **Software Engineer**, Misinformation Transparency *Menlo Park, CA*

Facebook

- Worked full-stack on Facebook/Instagram features to inform users about fact-checks for false information
- Designed and built a caching framework to handle the load of trillions of requests per day, saving ~$1MM annually in compute
- Enhanced data quality, leveraged logging to inform product decisions and discover anomalies
- Used static analysis to discover and remove dead code across the entire web codebase

May 2018-Aug. 2018, May 2017-Aug. 2017

Software Engineering Intern, Social Video Discovery — *Menlo Park, CA*
Software Engineering Intern, Messenger Monetization — *Menlo Park, CA*
Facebook

- Wrote Flux + React code with a GraphQL backend to build an interactive video editing UX
- Enhanced Facebook's video encoding infrastructure, resulting in an order-of-magnitude speedup in multiple product flows
- Worked full-stack on improving the creation flow for Click-to-Messenger advertisements

Selected technical projects

Dec. 2023-present **reagent**

- Designed and developed an open-source Web platform to facilitate rapid prototyping of AI-backed software for developers
- Deployed in the classroom, with tools for teachers to fund and assist students' experimentation
- Studied a classroom deployment in summer 2024, culminating in ITiCSE 2025 publication

May 2020-present **Language Transfer app**

- Designed, built, and shipped an open-source app for Language Transfer, a free series of language courses for beginners
- Developed and currently maintain backend infrastructure with Docker and AWS
- To date, **1,000,000+ installs, 10,000+ ratings, and 4.9 overall rating on Google Play**

Involvement and Recognition

Mar. 2025 **Outstanding Graduate Student Instructor Award**, UC Berkeley

Aug. 2022 **Berkeley Fellowship** awardee

June 2020 **Institute for Data, Democracy, & Politics**, George Washington University

- Invited to interview, included in Forum on Social Media Disinformation and Election Interference

Oct. 2020 (Speaker) **HexLabs**

- Invited guest speaker at HackGT 7 (2020)

Jan. 2018-May 2019 (Organizer),

- Assisted in organizing and running HexLabs events, including HackGT
- Developed curriculum for events, including Catalyst, a CS event for underserved HS students
- Developed and presented an educational React workshop at HackGT 5

Dec. 2018, May 2019 **Center for Teaching and Learning**, Georgia Institute of Technology

- Thank a Teacher award

Mar. 2016 **National Merit Scholar**